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Introduction

Welcome to the documentation for free travelling. This guide will help you get started with the basics and explore the more advanced features.

What is free travelling?

free travelling is a lightweight, open-source framework for building modern web applications. It provides a set of tools and conventions that make it easy to create fast, reliable, and maintainable software.

Key features

- Simple and intuitive API
- Built-in support for routing, templating, and data access
- Extensible plugin system
- Comprehensive test suite
- Detailed documentation and examples

Who is this guide for?

This guide is for developers who want to learn how to use free travelling to build web applications. It assumes you have a basic understanding of programming concepts and web development.

How to use this guide

You can read this guide from start to finish, or you can jump to the sections that are most relevant to you. Each section is self-contained and includes examples that you can try out on your own.

Installation

Prerequisites

Before you can install free travelling, you need to have the following software installed on your system:

- A supported operating system (Linux, macOS, or Windows)
- A compatible version of the runtime environment
- A package manager

Installing the package manager

If you don't already have a package manager installed, you can install one by following the instructions for your operating system.

On Linux

Open a terminal and run the following command:

```
curl -fsSL https://example.com/install.sh | sh
```

On macOS

Open a terminal and run the following command:

```
curl -fsSL https://example.com/install.sh | sh
```

On Windows

Open a PowerShell window and run the following command:

```
iwr -useb https://example.com/install.ps1 | iex
```

Installing free travelling

Once you have a package manager installed, you can install free travelling by running the following command:

```
pkg install free-travelling
```

Verifying the installation

To verify that free travelling was installed correctly, run the following command:

```
free-travelling --version
```

You should see the version number of free travelling printed to the console.

Updating free travelling

To update free travelling to the latest version, run the following command:

```
pkg update free-travelling
```

Uninstalling free travelling

To uninstall free travelling, run the following command:

```
pkg remove free-travelling
```

Configuration

Configuration files

free travelling uses a configuration file to store its settings. The configuration file is a plain text file in the TOML format.

Location of the configuration file

The configuration file is located in the following directory:

- On Linux and macOS: `~/.config/free-travelling/config.toml`
- On Windows: `%APPDATA%\free-travelling\config.toml`

Creating the configuration file

If the configuration file does not exist, you can create it by running the following command:

```
free-travelling config init
```

This will create a default configuration file with all of the available settings.

Configuration options

General

```
[general]
# The name of the application
name = "free travelling"

# The environment in which the application is running
# Possible values: development, production, test
environment = "development"

# The log level
# Possible values: debug, info, warn, error
log_level = "info"
```

Server

```
[server]
# The host to bind to
host = "127.0.0.1"

# The port to listen on
port = 8080

# The number of worker threads
workers = 4
```

Database

```
[database]
# The database driver
# Possible values: sqlite, postgres, mysql
driver = "sqlite"

# The database connection string
connection = "free-travelling.db"

# The maximum number of connections in the pool
max_connections = 10
```

Templating

```
[templating]
# The directory where templates are stored
directory = "templates"

# The file extension for templates
extension = "html"
```

Environment variables

You can also configure free travelling using environment variables. Environment variables take precedence over the configuration file.

Environment variable	Configuration option
`FREE_TRAVELLING_NAME`	`general.name`
`FREE_TRAVELLING_ENVIRONMENT`	`general.environment`
`FREE_TRAVELLING_LOG_LEVEL`	`general.log_level`
`FREE_TRAVELLING_HOST`	`server.host`
`FREE_TRAVELLING_PORT`	`server.port`
`FREE_TRAVELLING_WORKERS`	`server.workers`
`FREE_TRAVELLING_DATABASE_DRIVER`	`database.driver`
`FREE_TRAVELLING_DATABASE_CONNECTION`	`database.connection`
`FREE_TRAVELLING_DATABASE_MAX_CONNECTIONS`	`database.max_connections`
`FREE_TRAVELLING_TEMPLATING_DIRECTORY`	`templating.directory`
`FREE_TRAVELLING_TEMPLATING_EXTENSION`	`templating.extension`

Command-line options

You can also configure free travelling using command-line options. Command-line options take precedence over environment variables and the configuration file.

```
free-travelling --help
```

This will print a list of all of the available command-line options.

Usage

Starting the server

To start the server, run the following command:

```
free-travelling serve
```

This will start the server using the configuration from the configuration file, environment variables, and command-line options.

Stopping the server

To stop the server, press `Ctrl+C` in the terminal where the server is running.

Running in the background

To run the server in the background, you can use the `&` operator:

```
free-travelling serve &
```

You can also use a process manager like `systemd` or `supervisor` to manage the server process.

Logging

The server logs to the standard output and standard error streams. You can redirect these streams to a file:

```
free-travelling serve > server.log 2>&1 &
```

Health check

You can check the health of the server by sending a request to the `/health` endpoint:

```
curl http://127.0.0.1:8080/health
```

If the server is healthy, it will return a `200 OK` response with the body `OK`.

Metrics

The server exposes metrics in the Prometheus format at the `/metrics` endpoint:

```
curl http://127.0.0.1:8080/metrics
```

Examples

Hello, world!

Here is a simple example of a free travelling application that returns "Hello, world!" for every request:

```
use free_travelling::{App, Request, Response};

fn main() {
    let app = App::new();
    app.get("/", |_req: Request| {
        Response::text("Hello, world!")
    });
    app.run("127.0.0.1:8080");
}
```

Routing

free travelling supports routing requests to different handlers based on the URL path and the HTTP method.

```
use free_travelling::{App, Request, Response};

fn main() {
    let app = App::new();
    app.get("/", |_req: Request| {
        Response::text("Home")
    });
    app.get("/about", |_req: Request| {
        Response::text("About")
    });
    app.post("/submit", |_req: Request| {
        Response::text("Submitted")
    });
    app.run("127.0.0.1:8080");
}
```

Templating

free travelling includes a templating engine that you can use to render HTML pages.

```
use free_travelling::{App, Request, Response};
use free_travelling::templating::Context;

fn main() {
    let app = App::new();
    app.get("/", |_req: Request| {
        let mut ctx = Context::new();
        ctx.insert("name", "world");
        Response::render("hello.html", &ctx)
    });
    app.run("127.0.0.1:8080");
}
```

```
<!-- templates/hello.html -->
<h1>Hello, {{ name }}!</h1>
```

Database access

free travelling provides a simple interface for accessing databases.

```
use free_travelling::{App, Request, Response};
use free_travelling::database::Database;

fn main() {
    let app = App::new();
    app.get("/users", |_req: Request| {
        let db = Database::connect("free-travelling.db").unwrap();
        let users = db.query("SELECT name FROM users").unwrap();
        Response::text(users.join("\n"))
    });
    app.run("127.0.0.1:8080");
}
```

Middleware

free travelling supports middleware for processing requests and responses.

```
use free_travelling::{App, Request, Response};
use free_travelling::middleware::Logger;

fn main() {
    let app = App::new();
    app.middleware(Logger::new());
    app.get("/", |_req: Request| {
        Response::text("Hello, world!")
    });
    app.run("127.0.0.1:8080");
}
```

Static files

free travelling can serve static files from a directory.

```
use free_travelling::{App, Request, Response};
use free_travelling::static_files::StaticFiles;

fn main() {
    let app = App::new();
    app.mount("/static", StaticFiles::new("public"));
    app.get("/", |_req: Request| {
        Response::text("Hello, world!")
    });
    app.run("127.0.0.1:8080");
}
```

Troubleshooting

The server won't start

If the server won't start, check the following:

- Is another process using the same port? Try changing the port in the configuration file or using the `--port` command-line option.
- Is the configuration file valid? Run `free-travelling config check` to validate the configuration file.
- Are the database credentials correct? Check the `database.connection` setting in the configuration file.

The server is slow

If the server is slow, check the following:

- Is the database slow? Enable query logging and look for slow queries.
- Is the server running out of memory? Check the memory usage of the server process.
- Is the server running out of CPU? Check the CPU usage of the server process.

Getting help

If you can't find the answer to your problem in this guide, you can get help from the following resources:

- The free travelling issue tracker
- The free travelling mailing list
- The free travelling chat room

FAQ

Is free travelling production-ready?

free travelling is still in beta, but it is stable enough for production use. We recommend that you test your application thoroughly before deploying it to production.

What is the roadmap for free travelling?

The roadmap for free travelling is available on the project website. It includes the features that we plan to add in the next few releases.

How can I contribute to free travelling?

We welcome contributions to free travelling. You can contribute by reporting bugs, suggesting features, writing documentation, or submitting code. See the Contributing section for more information.

What is the license for free travelling?

free travelling is released under the MIT license. See the License section for more information.

Contributing

Reporting bugs

If you find a bug in free travelling, please report it to the issue tracker. Include the following information in your report:

- The version of free travelling you are using
- The operating system you are using
- The steps to reproduce the bug
- The expected behavior
- The actual behavior

Suggesting features

If you have a feature that you would like to see in free travelling, please suggest it to the issue tracker. Include the following information in your suggestion:

- A description of the feature
- The use case for the feature
- The benefits of the feature

Writing documentation

We welcome contributions to the documentation. You can contribute by fixing typos, clarifying confusing sections, or adding new sections.

Submitting code

We welcome contributions to the code. You can contribute by fixing bugs, implementing features, or improving the performance of the code.

Setting up the development environment

To set up the development environment, you need to have the following software installed on your system:

- A supported operating system (Linux, macOS, or Windows)
- A compatible version of the runtime environment
- A package manager
- A version control system

Building free travelling

To build free travelling, run the following command:

```
pkg build free-travelling
```

Running the tests

To run the tests, run the following command:

```
pkg test free-travelling
```

Submitting a pull request

To submit a pull request, you need to have a GitHub account. Fork the repository, make your changes, and submit a pull request.

License

free travelling is released under the MIT license.

```
MIT License

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```